

Phase 4:

Method Selection

Overview

In this phase, you will select one interpretable deep learning method from a curated list of recent research papers and conduct a thorough technical and practical analysis of its architecture, design choices, and computational behavior. The goal is to develop an understanding of how interpretable models are constructed, why certain components are used, and what trade-offs exist between performance, interpretability, and resource efficiency—especially in the context of medical image classification.

You will:

- Choose one paper from the provided list and dissect its proposed method.
- Analyze its architecture, hyperparameters, and design rationale.
- Compare it with at least one alternative approach.
- Quantify its computational footprint (memory, parameters, bottlenecks).
- Propose practical strategies to optimize its resource usage.

This phase emphasizes architectural literacy, critical comparison, and computational awareness—essential skills for deploying responsible and efficient AI systems in clinical settings.

Step 1: Method Selection and Description (50%)

1. Choose one of the following papers:
 1. This Looks Like That: Deep Learning for Interpretable Image Recognition
 2. Interpretable Image Recognition by Constructing Transparent Embedding Space
 3. Deformable ProtoPNet: An Interpretable Image Classifier Using Deformable Prototypes
 4. Interpretable Image Classification with Differentiable Prototypes Assignment
 5. Neural Prototype Trees for Interpretable Fine-grained Image Recognition
 6. ProtoPFormer: Concentrating on Prototypical Parts in Vision Transformers for Interpretable Image Recognition

7. Interpretable Image Classification with Adaptive Prototype-based Vision Transformers
 8. PIP-Net: Patch-Based Intuitive Prototypes for Interpretable Image Classification
2. Describe the Proposed Method:
 - Explain the key components of the model (e.g., feature extractor, prototype layer, routing mechanism, attention modules, interpretability head).
 - Provide a high-level flowchart of the full architecture and detailed sub-diagrams for critical modules (e.g., prototype matching, capsule routing, or prototype projection).
 - List the main hyperparameters (e.g., number of prototypes, capsule dimensions, attention heads) and discuss how each affects model performance, interpretability, and training stability.
 3. Justify Your Choice:
 - Compare your selected method with at least one other method from the list.
 - Discuss the advantages and disadvantages of both approaches in the context of your medical classification problem (from Phase 3). For example, you may consider interpretability, data efficiency, robustness, and computational demands.
 4. Anticipate Challenges. Predict potential challenges of deploying your chosen method to your problem.

Step 2: Computational and Resource Analysis (50%)

1. Compute and report the total number of trainable parameters in the model.
2. Using synthetic input tensors matching your dataset's dimensions:
 - a. Estimate GPU/CPU memory usage during training (including gradients, optimizer states, and activation buffers).
 - b. Estimate memory usage during inference (forward pass only).
3. Identify the most computationally expensive components (e.g., attention layers, prototype similarity search, dynamic routing in capsules). Explain why these components are costly (e.g., quadratic complexity, iterative optimization, large intermediate tensors).
4. Propose three concrete potential strategies to reduce computational or memory demands.

Submission Instruction

The students should upload their report/notebook there according to this structure on their repository in <https://github.com/UT-AI-Lab>. The name of the submission file for this phase should be "<Student Id>_Phase4".

```
|— student_name_1/  
| |— "<Student Id>_Phase4".ipynb    # Main Jupyter notebook (answers + results)  
| ...
```

 **Deadline for submission: 30 Aban.**